

WASP-52b

R-1806

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_190910 · created 2026-08-03T02:59:48+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458736.8053 +/- 0.005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.216 +/- 0.064
Transit depth (Rp/Rs)^2	4.65 +/- 2.76 [%]
Orbital Inclination (inc)	85.91 +/- 2.71
Airmass coefficient 1 (a1)	1.105 +/- 0.092
Airmass coefficient 2 (a2)	-0.06 +/- 0.55
Scatter in the residuals of the lightcurve fit is	8.58 %
Best Comparison Star	#1 - [342, 141]
Optimal Aperture	4.5
Optimal Annulus	14.55
Transit Duration (day)	0.076 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.216 ± 0.064 vs 0.1646 (deviation 0.68σ)
Duration fit vs published	0.076 d vs 0.0758 d (deviation 0.01σ)
Mid-time O–C	-11.9 min
Depth SNR	2.8
Residual scatter	8.58 %

Light curve

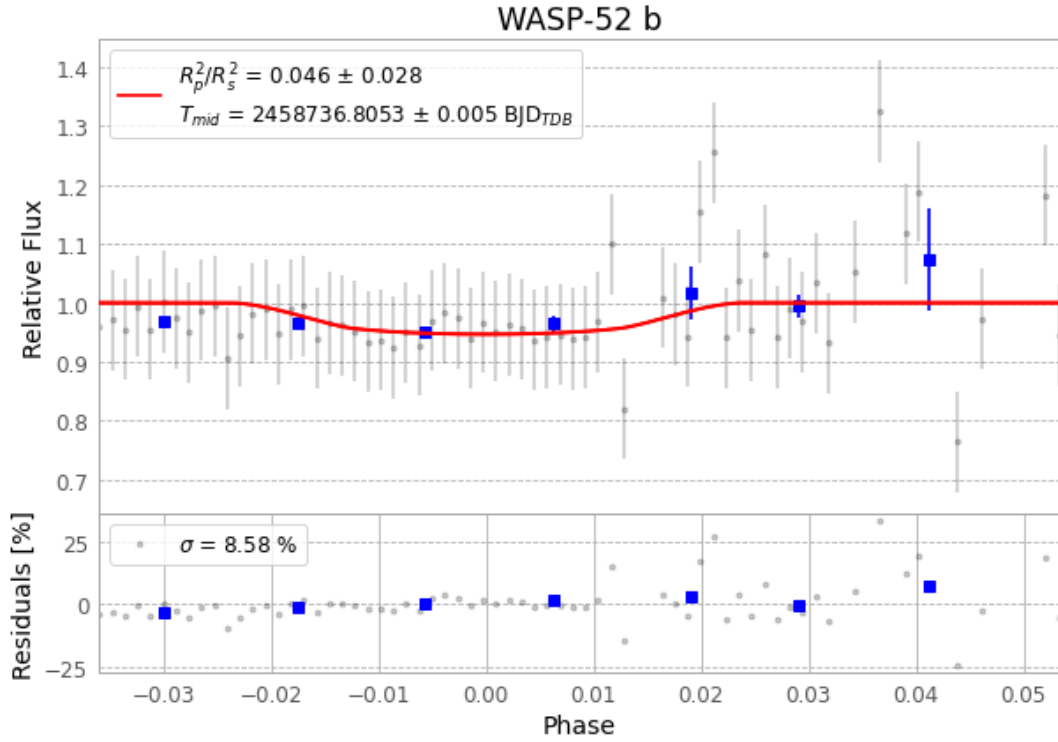


Figure 1 — FinalLightCurve_WASP-52 b_2019-09-09.png (SHA-256 4e6203522806aa91...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [159, 215], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 614321829924e103...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), WASP-52190910053912.FITS → WASP-52190910092412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-190910.log (88175c7993cb...), FinalLightCurve_WASP-52 b_2019-09-09.png (4e6203522806...), FinalLightCurve_WASP-52 b_2019-09-09.csv (244a93716b5f...)

Compute resources

Wall time 295.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 02:59Z.